Relational Algebra

Databases - Computer Science

1. Why an algebra

Relational algebra gives a small set of operators that take relations and return relations. Because the result is itself a relation, operators compose, and a query optimiser can rewrite one expression into an equivalent, cheaper one.

2. The core operators

Selection (sigma) keeps tuples satisfying a predicate. Projection (pi) keeps a subset of attributes, discarding duplicates. Union, difference and intersection apply to union-compatible relations. The Cartesian product pairs every tuple of one relation with every tuple of another. Rename (rho) supplies names where an expression would otherwise be ambiguous.

3. Joins

A theta join is a product followed by a selection; an equijoin restricts that predicate to equality. The natural join equates all attributes sharing a name and drops the duplicated columns. Outer joins retain unmatched tuples from one or both sides, padding the missing attributes with NULL.

4. Algebra and SQL

SQL is declarative - it states what is wanted, not how to compute it. The engine translates a query into an algebraic expression, then rewrites it: pushing selections below joins, for instance, shrinks intermediate results and is one of the most effective optimisations available.